



BASIC LINEAR ALGEBRA

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IIT Bombay

INTENDED AUDIENCE : B.Tech, B.Sc., M.Sc (maths) , BA/MA Economics/Statistics

PREREQUISITES : Core

INDUSTRY SUPPORT : Basic mathematics at school level undergraduate level

COURSE OUTLINE :

Linear Algebra is a basic course for students intended to go for Engineering studies and higher studies in Mathematics. We will cover the basic tools needed in these branches.

ABOUT INSTRUCTOR :

Prof. Inder K. Rana presently is an Emeritus Fellow at Department of mathematics, IIT Bombay. He has an experience of 36 years of teaching mathematics courses to undergraduate (B. Tech) and master's M.Sc. students at IIT Bombay. He has authored 4 books, namely, "Introduction to measure and Integration" American Mathematical Society, Graduate Studies in Mathematics Volume 45, 2000, "From Numbers to Analysis" World Scientific Press, 1998, "Calculus @IITB: Concepts and Examples, math4all, India, 2007" "From Geometry to Algebra: A course in Linear Algebra" math4all, India, 2007. He has won three awards, "C. L. Chandna Mathematics Award" for the year 2000 in recognition of distinguished and outstanding contributions to mathematics research and teaching. The award is given by 'SaraswatiVishvas Canada', "Excellence in Teaching" award for the year 2004 Awarded by IIT Bombay, based on the evaluations by students. "Aryabhata Award" 2012 All India Ramanujan Math Club, India, for teaching and work towards math education in India.

COURSE PLAN :

Week 1: Matrices and Matrix operations, REF

Week 2: Linear systems, Gauss Elimination and Inverse of a matrix

Week 3: $\mathbb{R}^n$; subspaces, linear independence, rank of a matrix

Week 4: Determinants, rank, invertibility

Week 5: Linear transformations, rank-nullity

Week 6: Inner product spaces, Gram-Schmidt process

Week 7: Eigenvalues and Eigenvectors

Week 8: Similarity, diagonalization and applications